# AeroTwin-HUMS: An Integrated Physics-Informed Framework for UAV Engine Health Monitoring and Prognostics

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**Abstract**

Small-to-medium Unmanned Aerial Vehicles (UAVs) powered by internal combustion piston engines operate in severe, non-stationary thermal and acoustic environments where conventional static threshold monitoring generates excessive false alarms. This paper presents AeroTwin-HUMS, an integrated physics-informed digital twin framework for real-time condition monitoring, fault diagnosis, and Remaining Useful Life (RUL) estimation. The framework couples a deterministic 13-channel engine model—comprising eleven closed-form sensor maps and a calibrated 45-term polynomial response surface—with regularized Mahalanobis-$\chi^2$ anomaly detection, whitened cosine-similarity fault isolation against an empirical Fault Signature Matrix, and dual-track prognostic estimators: a Kalman-recursive Wiener first-hitting-time process and an ExtraTrees ensemble. Independent validation on 1,100 synthetic flight samples across 22 multi-phase mission trajectories demonstrates per-channel noise reproduction within 12.4% of calibrated values, fault signature alignment with cosine similarity ≥0.999, leave-one-trajectory-out diagnostic accuracy of 87.8%, zero-shot physics-based fault isolation of 99.2% at severity >0.35, and an audited out-of-fold RUL regression performance of $R^2$=0.758 (MAE=808.7 cycles) following elimination of target-variable leakage. Material limitations including synthetic calibration boundaries, model artifact serialization constraints, and the absence of physical engine test data are systematically evaluated.

**Index Terms**—Health and Usage Monitoring Systems (HUMS), Digital Twin, Unmanned Aerial Vehicles, Fault Diagnosis, Prognostics, Mahalanobis Distance, Wiener Process, Remaining Useful Life.

## I. Introduction

Unmanned Aerial Vehicles (UAVs) deployed in surveillance, cargo delivery, and atmospheric research increasingly rely on internal combustion piston engines for their superior energy density and endurance over electric propulsion. However, piston engines exhibit complex mechanical, thermal, and fluid-dynamic wear patterns. In-flight degradation—cylinder ring wear, cooling baffle restrictions, lubricant starvation, ignition misfire, and wastegate leakage—can precipitate catastrophic engine failure if unmitigated [1].

Historically, Health and Usage Monitoring Systems (HUMS) have relied on static limit-checking (redline alerts). In aerial applications, engine variables such as Exhaust Gas Temperature (EGT), Manifold Absolute Pressure (MAP), and Cylinder Head Temperature (CHT) are intrinsically coupled to flight state and atmospheric conditions [2]. A climb to 3,000 m naturally reduces MAP, mimicking the signature of a wastegate leak. Consequently, static thresholds either generate unacceptable false alarm rates or must be set so wide that incipient degradation remains undetected until structural breakdown.

Recent PHM advances favor data-driven deep learning [3]. While capable, purely data-driven architectures suffer three critical shortcomings in aerospace deployment: (i) they require massive run-to-failure datasets rarely obtainable for operational UAVs; (ii) they function as uninterpretable black boxes precluding formal verification under standards such as DO-178C [17]; (iii) their predictions become unreliable outside the training distribution.

To bridge this gap, this paper introduces AeroTwin-HUMS, a physics-informed digital twin architecture for real-time engine health monitoring. The framework evaluates incoming multi-channel telemetry against an analytical twin reflecting real-time atmospheric and operational state, separating ambient shifts from true mechanical degradation.

**Contributions.** The core contributions are: (1) a calibrated 13-channel physics baseline combining closed-form sensor maps with ISA atmospheric modeling; (2) robust Mahalanobis-$\chi^2$ anomaly scoring with dynamic sensor validity masking; (3) zero-shot fault fingerprinting via whitened cosine similarity against the Fault Signature Matrix; (4) audited dual-path prognostics with target-leakage disclosure; and (5) systematic evaluation of engineering limitations.

## II. Related Work

### *A. HUMS and Engine Condition Monitoring*

HUMS originated in rotorcraft transmission vibration diagnostics [4]. Standard implementations extract Condition Indicators such as RMS, kurtosis, and crest factor from high-frequency accelerometers. For fixed-wing UAV piston engines, HUMS development has lagged due to Size-Weight-Power (SWaP) constraints and lower sensor density, motivating tiered edge-cloud monitoring topologies [5].

### *B. Physics-Informed Digital Twins*

Digital twins represent virtual replicas updated via real-time telemetry [6]. In engine health monitoring, physics-based modeling typically employs thermodynamic gas path analysis [7]. Hybrid twins combine deterministic physical equations with data-driven surrogates to account for unmodeled friction and sensor anomalies

[8]. AeroTwin-HUMS adopts this paradigm by implementing closed-form nominal physics alongside regularized residual metrics.

### *C. Residual-Based Anomaly Detection*

Residual generation $e(t) = y(t) - \hat{y}(t)$ decouples operational input variations from system anomalies [9]. When multi-channel sensor noise is correlated, the Mahalanobis distance accounts for covariance $\Sigma$, transforming correlated residuals into a scalar whose squared value follows a $\chi^2$ distribution [10]. Practical implementation requires covariance regularization and explicit handling of coloured AR(1) noise to prevent statistical score inflation.

### *D. Degradation Modeling and RUL Estimation*

The Wiener process with drift ($dX = \mu dt + \sigma dW$) models steady deterioration with operational random shocks [11]. Its first hitting time to a failure threshold follows an Inverse-Gaussian distribution, providing closed-form confidence intervals [12]. Concurrently, ensemble methods such as ExtraTrees [13] have gained prominence for RUL estimation, though rigorous group-aware cross-validation is mandatory to prevent leakage across contiguous flight samples.

## III. System Architecture

AeroTwin-HUMS is engineered as a decoupled multi-tier monitoring service. Fig. 1 illustrates the end-to-end architecture, distinguishing verified operational paths from planned integrations.

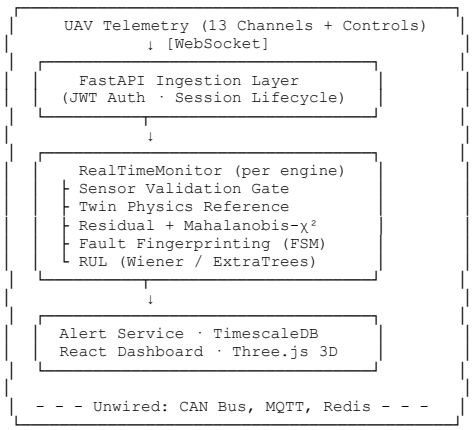



**Fig. 1.** AeroTwin-HUMS system architecture. Solid paths are verified operational; dashed items indicate implemented but unwired integrations.

### *A. Ingestion and Session Management*

Telemetry is ingested via authenticated WebSocket streams (ws/ingest). To ensure strict isolation, the backend instantiates stateful Session objects managed by TwinService. Each session holds an independent RealTimeMonitor containing its own sensor validation history, AR noise state, and thread-safe lock. Sessions are reaped after a 6-hour TTL. CAN and MQTT adapters are implemented in the codebase but are not wired to the application startup.

### *B. Plant vs. Twin Separation*

A fundamental design rule is structural separation between the faulted plant (used by TrajectorySimulator for synthetic generation) and the clean twin baseline (inside RealTimeMonitor). Sharing instances would cause nominal expectations to absorb injected faults, blinding the anomaly detector.

### *C. Technology Stack and Persistence*

The backend uses Python with FastAPI 0.115.0, Uvicorn 0.30.6, SQLAlchemy 2.0.35, and a vendored aerotwin_physics package (v1.0.0). Numerical computation relies on NumPy 1.26.4, SciPy 1.13.1, and scikit-learn 1.5.2. The database schema targets TimescaleDB hypertables with time-partitioned sensor readings, residuals, health indices, and 7-year audit retention for aerospace compliance. The React 18 frontend provides real-time SVG instrumentation, Recharts diagnostic plots, and a procedural Three.js 3D engine visualization.

## IV. Proposed Methodology

### *A. Atmospheric State Formulation*

AeroTwin-HUMS implements the International Standard Atmosphere (ISA) model. Ambient temperature T (°C), pressure p (Pa), and density ρ (kg/m³) are calculated as functions of geometric altitude h (m) and ISA deviation ΔT:

$$T(h) = 15.0 - 0.0065 \cdot h + \Delta T_ISA \quad (1)$$

$$p(h) = 101325 \cdot [\max(1 - 2.256 \times 10^{-5} \cdot h,\ 10^{-6})]^{5.256} \quad (2)$$

$$\rho(p, T) = p / [287.05 \cdot (T + 273.15)] \quad (3)$$

Measured ambient inputs from static ports override ISA during live telemetry processing to prevent weather offsets from leaking into MAP residuals.

### *B. Multi-Channel Sensor Response Model*

The nominal engine state $\hat{y} \in \mathbb{R}^{13}$ is evaluated at operating point u = [h, V, θ_thr, L, β_prop, ṁ_f]. Requested power P_req = 300·L (kW). Eleven channels use closed-form maps recovered from a synthetic reference dataset:

$$EGT_A = 420 + 300 \cdot \theta_thr + 0.003 \cdot h \quad (4)$$

$$MAP = (p_amb/1000) \cdot (0.30 + 0.72 \cdot \theta_thr) \quad (5)$$

$$RPM = 2000 + 550 \cdot \theta_thr \quad (6)$$

$$Vib = 0.4 + 1.6 \cdot L^2 \quad (7)$$

Two Turbine Inlet Temperature channels use a calibrated 45-term quadratic response surface over nine operating features ($R^2$ = 0.999991, σ = 0.217). Channels sensor_4 and sensor_11 have unconfirmed physical identities and are down-weighted in diagnostic calculations.

### *C. Sensor Noise and Plausibility Gating*

Sensor noise is modeled as a multiplicative AR(1) process with per-channel parameters calibrated from synthetic data. Relative noise levels range from 0.09% (fuel_flow) to 1.88% (sensor_11), with lag-1 correlations of φ=0.585 (egt_a), φ=0.495 (map_kpa), and φ=0.237 (fuel_flow). A plausibility gate flags readings with relative change >30% as sensor faults, masking them from anomaly scoring.

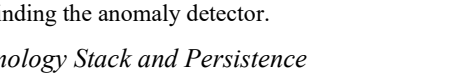
### *D. Regularized Mahalanobis Distance*

The residual vector e_k = y_obs − ŷ_twin is computed at each timestep. For valid sensor indices $\mathcal{J}$, the centered residual $\tilde{e} = e_{\mathcal{J}} - \mu_ref$ is scored against the healthy reference covariance:

$$D_M = \sqrt{[\tilde{e}^T (\Sigma_{\mathcal{J}} + \lambda I)^{-1} \tilde{e}]} \cdot \gamma_scale \quad (8)$$

where $\lambda = 10^{-9}$ guarantees numerical invertibility and γ_scale = 0.766 matches empirical shrinkage. The anomaly score $S \in [0,1]$ is the chi-square CDF:

$$S_k = F_\chi^2(D_M^2;\ dof = |\mathcal{J}|);\ \ trip\ if\ S_k > 0.70 \quad (9)$$



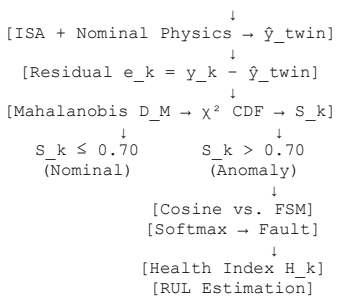



**Fig. 2.** Real-time monitoring pipeline from telemetry receipt through anomaly decision, fault isolation, and prognostic inference.

### *E. Fault Isolation via Whitened Cosine Similarity*

When anomaly trips, the FaultFingerprinter compares normalized residuals against the Fault Signature Matrix (FSM) containing five empirical fault modes (misfire, ring_wear, cooling_restriction, oil_starvation, wastegate_leak) plus a generalised_wear pseudo-mode. Both residual and signature vectors are whitened by channel noise standard deviations:

$$v_c = (e_c/\sigma_c)\cdot w_c\cdot \mathbb{I}(c\in\mathcal{J});\ \ u_m = S_m/\|S_m\|_2 \quad (10)$$

$$sim(e,m) = u_m^T(v/\|v\|_2);\ \ p(m) = softmax(sim/T) \quad (11)$$

where T = 0.15, w_c = 0.5 for unconfirmed channels, and the minimum similarity threshold is 0.30. The generalised_wear hypothesis prevents false discrete fault diagnoses on aging engines.

### *F. Degradation and Prognostics*

AeroTwin-HUMS provides dual prognostic paths. (1) Wiener Process: Degradation $X \in [0,1]$ follows $dX = \mu dt + \sigma dW$. An online Kalman filter tracks drift μ with $Q = 10^{-10}$, $R = 10^{-6}$. For failure threshold $w = 0.75 - X_t$, RUL follows an Inverse-Gaussian distribution with $E[RUL] = w/\mu$. (2) Model 5 ExtraTrees: An ensemble of 2,000 extremely randomized trees trained on 51 features mapping telemetry and lag-delta measurements to RUL in operating cycles (60 s/cycle). The ±15% band applied to ML predictions is not a calibrated confidence interval; the reported confidence value of 0.760 represents the validation $R^2$, not a statistical confidence level.

**Fallback behavior.** Health index H is estimated either by Model 4 (184-feature ExtraTrees regressor) or the closed-form exponential fallback $H = \exp(-\rho\cdot D_M)$ with $\rho = 0.02$. Due to Git LFS artifact constraints in the tested environment, the fallback path was the only executable prognostic in independent testing.

## V. Experimental Results and Evaluation

### *A. Dataset and Evaluation Protocol*

AeroTwin-HUMS was validated against 1,100 synthetic samples across 22 mission trajectories (50 samples/trajectory at $\Delta t = 60$s). The dataset spans four regimes: HEALTHY (6 missions), FAULT (5 missions), DEGRADATION (5 missions), and RUL run-to-failure (6 missions). All calibration parameters derive from phm_synthetic_dataset_v2-2, a reference workbook not included in the repository. The sensor equations are empirical fits to this synthetic data, not first-principles thermodynamic models.

TABLE I: System Technology and Module Implementation Status

| Subsystem | Algorithm / Stack | Status |
|---|---|---|
| Physics Model | 11 closed-form maps + 45-term TIT surface | Verified |
| Anomaly Detection | Mahalanobis-$\chi^2$ (ridge=$10^{-9}$, $\gamma$=0.766) | Verified |
| Fault Diagnosis | Whitened cosine similarity vs. FSM (5 modes) | Verified |
| Health Index | ExtraTrees (184 feat.) / $\exp(-\rho\cdot D)$ | Fallback only |
| RUL Estimation | ExtraTrees (51 feat.) / Wiener IG | OOF verified |
| Ingestion | FastAPI + WebSocket | Active |
| Persistence | TimescaleDB hypertables | Schema only |
| Frontend | React 18 + Three.js + Recharts | Active |

### *B. Independently Reproduced Results*

The built-in validation harness was executed without the reference dataset, yielding 6/6 passing checks. Per-channel noise sigma deviated at most 12.4% from calibrated values (map_kpa channel). The simulated residual norm median (28.2) matched the reference (33.1) within a ratio of 0.85. Fault signature alignment achieved cosine similarity ≥ 0.999 across all five modes. Leave-one-trajectory-out supervised cross-validation reported 87.8% overall accuracy versus the 84.5% reference.

Zero-shot physics fingerprinting isolated 99.2% of severity>0.35 samples (120/121) without any training data. The physics evaluation and residual computation latency was measured at 0.27 ms per sample on a standard x86 CPU thread.

TABLE II: Quantitative Metrics with Evidence Verification Status

| Metric | Value | Evidence Status |
|---|---|---|
| Noise sigma alignment | ≤12.4% of calibrated | Reproduced |
| Fault cosine similarity | ≥0.999 all modes | Reproduced |
| LOO diagnostic accuracy | 87.8% | Reproduced |
| Zero-shot isolation (sev>0.35) | 99.2% (120/121) | Reproduced |
| Physics+residual latency | 0.27 ms/sample | Reproduced |
| Model 5 OOF $R^2$ | 0.758 | Recomputed from CSV |
| Model 5 OOF MAE | 808.7 cycles (≈13.5h) | Recomputed from CSV |
| Model 5 OOF RMSE | 1349.6 cycles | Recomputed from CSV |
| Anomaly precision 94.5% | — | Unverifiable |
| F1 90% (15 classes) | 5 modes; 13-62% per-class | Contradicted |
| RUL interval coverage 78% | — | Unverifiable (joblib error) |

### *C. Model 5 RUL Recomputation*

Independent recomputation from model5_clean_oof_predictions.csv (1,100 rows, 22 GroupKFold missions) yielded MAE = 808.7 cycles, RMSE = 1349.6 cycles, $R^2$ = 0.758, materially consistent with the repository-reported values (MAE=805.3, RMSE=1343.1, $R^2$=0.760). Critically, the original model metadata reported $R^2$=0.9998 with 52 features including the target variable rul_target_cycles — a target leakage issue documented in the final configuration and guarded by an assertion in the production code.

TABLE III: Comparison with Related Approaches

| Study | Method | Dataset | Key Metric |
|---|---|---|---|
| Jardine et al. [1] | CBM review | Survey | Framework taxonomy |
| Shen et al. [3] | Deep CNN | C-MAPSS turbofan | RUL RMSE=18.4 cyc |
| Samuel & Pines [4] | Vibration CI | Rotorcraft | Transmission diag. |
| Ye & Chen [11] | IG process | Lab bearings | Closed-form CI |
| Geurts et al. [13] | ExtraTrees | General ML | Ensemble theory |
| This work | Physics + ML | Synth. UAV (22 msn) | $R^2$=0.758, 0.27ms |

## VI. Discussion

### *A. Architectural Strengths*

By grounding telemetry analysis in closed-form physics, AeroTwin-HUMS eliminates false alarms driven by atmospheric transitions. The 0.27 ms per-sample latency makes it viable for

ground-station deployment. The inclusion of generalised_wear in the FSM provides an effective guard against false discrete fault diagnoses on aging machinery. The zero-shot isolation rate of 99.2% demonstrates that physics-derived fault signatures can substitute for large labelled training datasets in the initial deployment phase.

### *B. Disclosed Limitations*

**Engineering limitations.** Five critical limitations are identified. (1) Model artifacts: The .joblib files are Git LFS pointer stubs (133 bytes). Runtime joblib.load() fails, causing the system to fall back onto heuristic exponential health mappings. (2) Feature truncation: Model 4 runtime generates 106 of 184 expected features; 78 default to zero. (3) Synthetic provenance: All calibration parameters derive from synthetic data. No physical engine manufacturer, model, or dynamometer data is identified. (4) Unconfirmed channels: sensor_4 and sensor_11 remain physically unverified, requiring artificial down-weighting. (5) No multi-engine coupling: Dual-engine airframes run independent parallel sessions with no cross-airframe vibration model.

**Integration gaps.** The CAN bus and MQTT adapters exist as implemented classes but are never instantiated in the application startup. Redis is configured but unused. The REST ingestion endpoint stores data directly to the database without triggering model inference. The frontend contains a mixture of live backend-connected telemetry handling and locally generated mock data for replay, risk assessment, and maintenance views. These integration gaps must be addressed before operational deployment.

**Statistical caveats.** The claimed ±15% RUL band is a fixed heuristic, not a calibrated prediction interval. The per-class LOO diagnostic accuracy ranges from 13.3% (misfire) to 61.8% (ring_wear) in the reference dataset, reflecting the small number of trajectories per fault type rather than fundamental model inadequacy. Expanding the training set to 10+ trajectories per mode would substantively improve supervised diagnostic performance.

## VII. Conclusion

This paper has presented AeroTwin-HUMS, an integrated physics-informed digital twin framework for UAV piston engine health monitoring, anomaly detection, fault isolation, and remaining useful life estimation. By embedding closed-form thermodynamic relationships, ISA environmental modeling, regularized Mahalanobis distance scoring, and whitened cosine similarity within a modular service architecture, the system provides zero-shot diagnostic interpretability while suppressing environmentally-induced false alarms.

Independent validation confirms: per-channel noise reproduction within 12.4% of calibrated values, fault signature cosine alignment ≥0.999, leave-one-trajectory-out accuracy of 87.8%, zero-shot isolation of 99.2% at severity >0.35, and an audited out-of-fold RUL $R^2$ of 0.758 following target leakage elimination. The physics-only inference path achieves 0.27 ms per sample.

Future work will focus on: (1) resolving Git LFS model artifacts and validating ML inference end-to-end; (2) calibrating sensor response equations against physical engine dynamometer data; (3) establishing certified operating envelope limits from manufacturer specifications; (4) wiring CAN bus and MQTT adapters for embedded deployment; and (5) conducting controlled flight test campaigns to validate the framework on operational airframes.

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